Felipe Portugal

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linkedin.com/in/felipe-portugal-151a75225 | Trilingual: French | English | Spanish

Focus: Software Engineering student: full-stack web and mobile development, backend, embedded systems, IoT and robotics.
Availability: 4- or 8-month internships | full-time | active student, returning to studies after each internship.

Education

Bachelor of Software Engineering (B.Eng.)
École de technologie supérieure (ÉTS), Montreal | since 2024 | expected graduation: May 2029

DEC (Diploma of College Studies) in Computer Science Technology
Cégep de Sept-Îles | graduated in 2024

Technical Skills

Languages: C#, Java, Python, TypeScript, JavaScript, PHP, SQL, PL/SQL, C, C++, Rust, Bash, PowerShell, VBA
Web and .NET: ASP.NET Core (MVC, Blazor, Web API), Entity Framework, REST APIs, HTML, CSS, Bootstrap, jQuery, AJAX
Mobile: React Native, Expo, Android
Data: SQL Server, PostgreSQL, Oracle, MySQL, MariaDB, SQLite, migrations, database modelling
Testing and Quality: pytest, JaCoCo, unit, integration and black-box testing, code coverage, reverse engineering
DevOps / Systems: Docker, Git, Gitea, GitLab, Azure DevOps, Linux, Windows Server (Active Directory, GPO), IIS, VMware ESXi, Apache, Zabbix, Grafana, Maven, Agile/Scrum
Embedded / IoT / Robotics: Spot SDK, Raspberry Pi, Arduino, MQTT, Bluetooth LE, LoRaWAN, GPIO/I2C/SPI/UART, OpenCV, Pillow, NumPy
Design and Tools: MVC architecture, design patterns, UML, JavaFX, Odoo, Wireshark, Cisco Packet Tracer, OCS Inventory, GLPI

Professional Experience

Institut technologique de maintenance industrielle (ITMI)
College technology transfer centre in industrial maintenance | Sept-Îles, Quebec | January 2025 – September 2025

Assistant Developer, Industrial Data Visualization | *May 2025 – September 2025*
Developer Intern, Industrial Data Visualization | *January 2025 – May 2025*

Shared contributions across both roles

- Independently developed the client-side mobile application for a parking management system deployed for all Cégep de Sept-Îles staff and a few external users, using React Native, TypeScript and Expo.
- Developed the administrator web interface for the same system: pages, forms and views.
- Developed industrial data visualizations with PHP and Grafana.
- Developed features in a web application for inventory and contract management.
- Contributed to REST APIs and backend components in PHP for Nomad, ITMI's internal framework, with PostgreSQL (connection, queries, migrations), in Docker-based projects.
- Designed SQL Server schemas and queries for web platforms in the development phase.
- Implemented basic authentication, Microsoft authentication and Stripe integration.
- Wrote automated client- and server-side tests in Python with pytest, including load tests.
- Worked in Agile/Scrum with pull requests, code reviews and a branching strategy in Gitea, tracking work in Azure DevOps Boards.
- Maintained applications in production, trained staff, and worked with clients and stakeholders.

Technical Advisor – Embedded Systems
Summer 2024 | ITMI, Sept-Îles, Quebec

- Developed a functional prototype for Boston Dynamics Spot in Python with the Spot SDK: selecting autonomous missions and recording new ones.
- Automated in-mission photo capture, upload to a Linux server and processing with payload data (OpenCV, Pillow, NumPy).
- Developed outdoor-sensor data collection in Rust and visualized the data in Grafana.
- Trained staff and wrote the technical documentation for the Spot projects.

Research Assistant
Winter 2024 | ITMI, Sept-Îles, Quebec

- Developed a Python proof of concept with the Spot SDK for Boston Dynamics Spot, including testing and validation; worked with various sensors and payloads; explored C++ with the Spot SDK.

IT Operator
Summer 2023 (internship), Fall 2023 (employment) | CISSS de la Côte-Nord (regional health network), Sept-Îles, Quebec

- Replaced and configured workstations and laptops; replaced components, disassembled and reassembled equipment, and installed cabling.
- Deployed updates and installed applications on Windows; configured network hardware; provided hardware and software support and assisted technicians.

Technical Projects — ÉTS

- **Software Testing and Quality — Java Application** *(2025, team, LOG240)*: reverse engineering, impact analysis of a change request, black-box testing (equivalence classes, boundary values), unit and integration tests for the data access layer (99% coverage, JaCoCo), defect tracking in GitLab.
- **Object-Oriented Design — JavaFX Application** *(2025, team, LOG121)*: image manipulation (zoom, translation, undo, save) using the MVC architecture and design patterns.
- **Software Analysis and Design** *(2026, LOG210)*: UML domain models, operation contracts and use-case realizations applying GRASP patterns.

Projects and Coursework — DEC, Cégep de Sept-Îles

- **ASP.NET Core Web Applications** *(2023–2024)*: MVC applications (controllers, views, CRUD) with Entity Framework and SQL Server; Blazor application managing students, courses and enrolments; Web API REST service consumed by a Blazor client and deployed on IIS.
- **Websites with Authentication — PHP and MySQL** *(2024)*: hashed passwords, sessions, roles, PDO prepared statements and XSS protection; OAuth (Google, Facebook, GitHub), QR-code two-factor authentication, email activation, reCAPTCHA.
- **Connected Spot Robotic Platform** *(2023, team of two)*: Autowalk mission during which the thermal camera locates the hottest heat source and the robot moves toward it; collection of audio, images and sensor data; Docker container (docker-compose) run over SSH on Spot and a Bash script launched during the mission.
- **Android Monitoring System** *(2024, individual)*: Android application (Java), Raspberry Pi (Python), Arduino (GPIO, I2C, SPI, UART, RTC module); requirements specification, schedule, client meetings and deliverable validation.
- **Motion Detection — Android and Raspberry Pi** *(2024)*: Java application with MQTT (Eclipse Paho), SQLite history and video; Python script on Raspberry Pi (motion sensor, camera).
- **Programmable LED Panel** *(2024, individual)*: Arduino MKR1010, light and temperature sensors, MQTT communication.
- **Connected Objects** *(2023)*: Arduino MKR labs: MQTT (Mosquitto), Bluetooth LE, LoRaWAN (The Things Network), sensors and serial links.
- **Server Infrastructure, Asset Management and Monitoring** *(2023)*: VMware ESXi, Ubuntu Server 22.04 LTS, Ubuntu and Windows 10 clients, DHCP; OCS Inventory, GLPI and Zabbix with Apache, MariaDB and PHP.
- **Oracle Databases** *(2023)*: PL/SQL: stored procedures and functions, cursors, exception handling; queries with joins, subqueries and aggregation; data modelling and normalization.
- **Client-Side Programming** *(2023)*: multipage website in HTML, CSS and Bootstrap; DOM manipulation and real-time search with jQuery and AJAX.
- **Servers and Networking** *(2022–2023)*: Windows Server (Active Directory, GPO, IIS, RAID); courses based on the Cisco CCNA curriculum (OSPF, VLANs, IPsec VPN, port security).
- **IT Security** *(2023, team of three)*: risk analysis and security plan; analysis of the Log4j vulnerability.
- **Battleship** *(2023)*: Java game with JavaFX, structured and built with Maven.

Involvement

- **CAPRA — Member of the Software Team** *(Fall 2024, student club, ÉTS)*: research on components for robotic platforms, hardware performance and compatibility testing.